

13. Two important Theorems

I. The moment generating function

def: Let X be a random variable. The moment generating function of X is $M(t) = E[e^{tX}]$.

We assume that $\exists \epsilon, K > 0$ such that $\forall t \in (-\epsilon, \epsilon)$, $|M(t)| < K$.
Clearly, $M(0) = E[e^{0 \cdot X}] = E[1] = 1$.

Prop: $\forall n \geq 1$: $M^{(n)}(0) = E[X^n]$, where $M^{(n)}$ denotes the n^{th} derivative of M .

proof: $M'(t) = \frac{d}{dt} E[e^{tX}] = E\left[\frac{d}{dt} e^{tX}\right] = E[X e^{tX}] \Rightarrow M'(0) = E[X]$

we assumed that $M(t)$ is bounded around 0, so the dominated convergence thm. applies

$$M^{(n)}(t) = \frac{d}{dt} M^{(n-1)}(t) = E\left[\frac{d}{dt} (X^{n-1} e^{tX})\right] = E[X^n e^{tX}] \Rightarrow M^{(n)}(0) = E[X^n]$$
$$= E[X^{n-1} e^{tX}]$$

Note: $E[X^n]$ is called the n^{th} moment of X , hence the name moment generator for $M(t)$. □

Example: $X \sim \text{Bin}(n, p)$

$$M(t) = E[e^{tX}] = \sum_{k=0}^n e^{tk} \binom{n}{k} p^k (1-p)^{n-k} = (pe^t + 1-p)^n$$

binomial thm.

$$M(0) = (p + 1-p)^n = 1$$

$$M'(t) = n(pe^t + 1-p)^{n-1} pe^t \Rightarrow M'(0) = E[X] = np$$

$$M''(t) = n(n-1)(pe^t + 1-p)^{n-2} p^2 e^{2t} + n(pe^t + 1-p)^{n-1} pe^t \Rightarrow M''(0) = E[X^2] = n(n-1)p^2 + np$$
$$\Rightarrow \underline{D^2[X]} = E[X^2] - E^2[X] = n(n-1)p^2 + np - n^2p^2 = \underline{np(1-p)}$$

Example: $X \sim \text{Poi}(\lambda)$

$$M(t) = E[e^{tX}] = \sum_{k=0}^{\infty} e^{tk} \frac{e^{-\lambda} \lambda^k}{k!} = e^{(e^t - 1)\lambda}$$

Example: $X \sim \text{Exp}(\lambda)$

$$M(t) = E[e^{tX}] = \int_0^{\infty} e^{tx} \lambda e^{-\lambda x} dx = \begin{cases} \infty, & \lambda \leq t \\ \frac{\lambda}{\lambda - t} \int_0^{\infty} (\lambda - t) e^{-(\lambda - t)x} dx = \frac{\lambda}{\lambda - t}, & t < \lambda \end{cases}$$

Example: $Y \sim N(0, 1)$

$$M_Y(t) = E[e^{tY}] = \int_{-\infty}^{\infty} e^{ty} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{(y-t)^2}{2} + \frac{t^2}{2}} dy = e^{\frac{t^2}{2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{(y-t)^2}{2}} dy = e^{\frac{t^2}{2}}$$

$X \sim N(\mu, \sigma^2)$

$$M_X(t) = E[e^{tX}] = E[e^{t\sigma Y + t\mu}] = e^{t\mu} E[e^{t\sigma Y}] = e^{t\mu} e^{\frac{t^2 \sigma^2}{2}}$$

density of $N(t, 1)$

Thm: Let X, Y be random variables and suppose that

$$\exists \varepsilon > 0, \forall t \in (-\varepsilon, \varepsilon), M_X(t) = M_Y(t).$$

Then the distribution of X and Y are equal.

$$(\forall z \in \mathbb{R}, F_X(z) = P(X \leq z) = P(Y \leq z) = F_Y(z))$$

Prop: If X and Y are independent random variables, then $M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$

proof: $M_{X+Y}(t) = E[e^{t(X+Y)}] = E[e^{tX} e^{tY}] = \underset{\text{indep.}}{E[e^{tX}] \cdot E[e^{tY}]} = M_X(t) M_Y(t) \quad \square$

Cor: Let $X \sim \text{Bin}(n, p), Y \sim \text{Bin}(m, p)$ be independent random variables, then

$$X+Y \sim \text{Bin}(n+m, p)$$

proof: $M_{X+Y}(t) = M_X(t) \cdot M_Y(t) = (pe^t + 1-p)^n (pe^t + 1-p)^m = (pe^t + 1-p)^{n+m} \xrightarrow{\text{thm.}} X+Y \sim \text{Bin}(n+m, p) \quad \square$

Cor: Let $X \sim \text{Poi}(\lambda), Y \sim \text{Poi}(\mu)$ be independent random variables, then

$$X+Y \sim \text{Poi}(\lambda+\mu)$$

proof: $M_{X+Y}(t) = M_X(t) \cdot M_Y(t) = e^{\lambda(e^t-1)} e^{\mu(e^t-1)} = e^{(\lambda+\mu)(e^t-1)} \xrightarrow{\text{thm.}} X+Y \sim \text{Poi}(\lambda+\mu) \quad \square$

Cor: $X \sim N(\mu_1, \sigma_1^2), Y \sim N(\mu_2, \sigma_2^2)$ are independent, then

$$X+Y \sim N(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2)$$

proof: $M_{X+Y}(t) = M_X(t) M_Y(t) = e^{\mu_1 t + \frac{\sigma_1^2 t^2}{2}} e^{\mu_2 t + \frac{\sigma_2^2 t^2}{2}} = e^{(\mu_1+\mu_2)t + \frac{(\sigma_1^2+\sigma_2^2)t^2}{2}} \xrightarrow{\text{thm.}} X+Y \sim N(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2) \quad \square$

def: Let $X_1, \dots, X_n$ be random variables. We call the function

$$M(t_1, \dots, t_n) = E[e^{t_1 X_1 + \dots + t_n X_n}] \text{ the joint moment generating function of } X_1, \dots, X_n.$$

Note: $M(0, \dots, 0, \underset{i\text{th variable}}{t_i}, 0, \dots, 0) = E[e^{0 \cdot X_1 + \dots + 0 \cdot X_{i-1} + t_i X_i + 0 \cdot X_{i+1} + \dots + 0 \cdot X_n}] = E[e^{t_i X_i}] = M_{X_i}(t_i)$

Thm: $X_1, \dots, X_n$ are independent random variables if and only if

$$\exists \varepsilon > 0, \forall t_1, \dots, t_n \in (-\varepsilon, \varepsilon), M(t_1, \dots, t_n) = M_{X_1}(t_1) \cdot \dots \cdot M_{X_n}(t_n)$$

Example: (Revisiting the post office)

The customers arrive to the post office according to a Poisson process with rate λ . A customer is male with probability p and female with probability $1-p$, independently.

X = number of male customers in a day

Y = number of female customers in a day

Show that X, Y are independent and $X \sim \text{Poi}(p\lambda), Y \sim \text{Poi}((1-p)\lambda)$

According to the exercise $X+Y \sim \text{Poi}(\lambda)$, $X|X+Y \sim \text{Bin}(X+Y, p)$.

We already solved this exercise earlier using the joint distribution of (X, Y) . Now we solve it using moment generating functions.

$$\begin{aligned} \mathbb{E}[e^{tX}] &\stackrel{\text{tower-rule}}{=} \mathbb{E}[\mathbb{E}[e^{tX}|X+Y]] \stackrel{X|X+Y \sim \text{Bin}(X+Y, p)}{=} \mathbb{E}[(pe^t + 1-p)^{X+Y}] \stackrel{X+Y \sim \text{Poi}(\lambda)}{=} \mathbb{E}[e^{(X+Y)\ln(pe^t + 1-p)}] \\ &= e^{\lambda(e^{\ln(pe^t + 1-p)} - 1)} = e^{\lambda(pe^t + 1-p - 1)} = e^{\lambda p(e^t - 1)} \stackrel{\text{thm.}}{\Rightarrow} X \sim \text{Poi}(\lambda p) \end{aligned}$$

Similarly, $M_Y(t) = \mathbb{E}[e^{tY}] = e^{\lambda(1-p)(e^t - 1)} \Rightarrow Y \sim \text{Poi}(\lambda(1-p))$

$$\begin{aligned} \underline{M_{X,Y}(t,s)} &= \mathbb{E}[e^{tX+sY}] = \mathbb{E}[e^{t(X+Y)+(s-t)Y}] \stackrel{\text{tower-rule}}{=} \mathbb{E}[\mathbb{E}[e^{t(X+Y)+(s-t)Y}|X+Y]] = \\ &= \mathbb{E}[e^{t(X+Y)} \mathbb{E}[e^{(s-t)Y}|X+Y]] \stackrel{Y|X+Y \sim \text{Bin}(X+Y, 1-p)}{=} \mathbb{E}[e^{t(X+Y)} ((1-p)e^{s-t} + p)^{X+Y}] = \\ &= \mathbb{E}[e^{(X+Y)(\ln((1-p)e^{s-t} + p) + t)}] \stackrel{X+Y \sim \text{Poi}(\lambda)}{=} e^{\lambda(e^{t((1-p)e^{s-t} + p)} - 1)} = e^{\lambda(1-p)(e^s - 1)} e^{\lambda p(e^t - 1)} = \underline{M_X(t) M_Y(s)} \\ &\Rightarrow X \text{ and } Y \text{ are independent.} \end{aligned}$$

II. The weak law of large numbers

Propo (Markov's inequality) Let X be a non-negative random variable ($P(X \geq 0) = 1$). Then for all $a > 0$

$$P(X > a) \leq \frac{\mathbb{E}[X]}{a}$$

proof: $\mathbb{E}[X] \geq \mathbb{E}[X \cdot \mathbb{1}_{\{X > a\}}] \geq \mathbb{E}[a \cdot \mathbb{1}_{\{X > a\}}] = a \mathbb{E}[\mathbb{1}_{\{X > a\}}] = a P(X > a) \quad \square$

Propo (Chebisev's inequality) Let X be a random variable with $\mathbb{E}[X] = \mu$, $\mathbb{D}^2[X] = \sigma^2$. Then

$$P(|X - \mu| > b) \leq \frac{\sigma^2}{b^2}, \quad \forall b > 0.$$

proof: $P(|X - \mu| > b) = P((X - \mu)^2 > b^2) \stackrel{\text{Markov}}{\leq} \frac{\mathbb{E}[(X - \mu)^2]}{b^2} = \frac{\sigma^2}{b^2} \quad \square$

Example: As Chebisev's inequality works in general for any X with finite expectation, it is not surprising that the upper bound it gives is rather weak.

$$X \sim N(\mu, \sigma^2)$$

with Chebisev: $P(|X - \mu| > 2\sigma) \leq \frac{\sigma^2}{(2\sigma)^2} = \frac{1}{4} = 0.25$

real value: $\underline{P(|X - \mu| > 2\sigma)} = \underline{P\left(\frac{X - \mu}{\sigma} > 2\right) + P\left(\frac{X - \mu}{\sigma} < -2\right)} = 2 - 2\Phi(2) \approx \underline{0.0456}$

Prop. (Turbo-Markov's inequality) Let X be a random variable, for $c \in \mathbb{R}$

$$P(X > c) \leq \frac{M_X(t)}{e^{ct}}, \quad \forall t > 0$$

proof.

$$\forall t > 0: P(X > c) = P(e^{tX} > e^{tc}) \leq \underbrace{\frac{E[e^{tX}]}{e^{tc}}}_{\text{Markov}} = \frac{M_X(t)}{e^{tc}} \quad \square$$

Thm. (WLLN - Weak Law of Large Numbers)

Let $X_1, X_2, \dots$ be a sequence of iid random variables with expectation $E[X_1] = \mu < \infty$ and variance $D^2[X_1] = \sigma^2 < \infty$. Then

$$\forall \varepsilon > 0: \lim_{n \rightarrow \infty} P\left(\left|\frac{X_1 + \dots + X_n}{n} - \mu\right| > \varepsilon\right) = 0.$$

Note. $\mu < \infty$ is already enough for this statement to hold.

proof. Let $\bar{X}_n = \frac{X_1 + \dots + X_n}{n}$, then $E[\bar{X}_n] = \mu$, $D^2[\bar{X}_n] = \frac{1}{n^2} \sum_{i=1}^n D^2[X_i] = \frac{\sigma^2}{n}$
↑
indep.

By Chebisev's inequality

$$P\left(\left|\frac{X_1 + \dots + X_n}{n} - \mu\right| > \varepsilon\right) = P(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{D^2[\bar{X}_n]}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \xrightarrow[n \rightarrow \infty]{} 0 \quad \square$$

III.] The central limit theorem

Prop. Let $(Z_n)_{n \geq 1}$ be a sequence of random variables. Let Z be a random variable and suppose that $\exists \varepsilon > 0, \forall t \in (-\varepsilon, \varepsilon)$,

$M_{Z_n}(t)$ is uniformly bounded on $(-\varepsilon, \varepsilon)$, and $M_{Z_n}(t) \xrightarrow[n \rightarrow \infty]{} M_Z(t)$.

Then $F_{Z_n}(x) \xrightarrow[n \rightarrow \infty]{} F_Z(x) \quad \forall x \in \mathbb{R}$, if F_Z is continuous at x .

(F_Z : distribution function of Z , M_Z : moment generating function of Z)

Thm. (CLT - Central Limit Theorem)

Let $X_1, X_2, \dots$ be iid. random variables such that

$\exists \varepsilon, K > 0, \forall t \in (-\varepsilon, \varepsilon), \forall i \geq 1, M_{X_i}(t) < K$.

Then $\lim_{n \rightarrow \infty} P\left(b \leq \frac{X_1 + \dots + X_n - n\mu}{\sqrt{n}\sigma} < a\right) = \Phi(a) - \Phi(b)$,

where $\mu = E[X_1]$, $\sigma^2 = D^2[X_1]$ and Φ is the standard normal distribution function.

Note: In fact, it is enough to only require $\mu < \infty$, $\sigma^2 < \infty$.

The CLT is a generalized form of the De Moivre-Laplace theorem.

proof:

Let $Y_i = \frac{X_i - \mu}{\sigma}$, then $E[Y_i] = 0$ and $D^2[Y_i] = 1$.

We want to prove that $P\left(\frac{Y_1 + \dots + Y_n}{\sqrt{n}} < a\right) \xrightarrow[n \rightarrow \infty]{} \Phi(a)$.

According to our last proposition, it is enough to show that

$$M_{\frac{Y_1 + \dots + Y_n}{\sqrt{n}}}(t) \xrightarrow[n \rightarrow \infty]{} e^{t^2/2}.$$

$$M_{\frac{Y_1 + \dots + Y_n}{\sqrt{n}}}(t) = E\left[e^{t \frac{Y_1 + \dots + Y_n}{\sqrt{n}}}\right] = \underset{\text{indep.}}{E\left[e^{t \frac{Y_1}{\sqrt{n}}}\right] \cdot \dots \cdot E\left[e^{t \frac{Y_n}{\sqrt{n}}}\right]} = \underset{\text{iid.}}{E\left[e^{t \frac{Y_1}{\sqrt{n}}}\right]^n} = \left(M_{Y_1}\left(\frac{t}{\sqrt{n}}\right)\right)^n$$

$$= \left(1 + \frac{t^2}{2n} + O\left(\frac{t^3}{n\sqrt{n}}\right)\right)^n \xrightarrow[n \rightarrow \infty]{} e^{\lim_{n \rightarrow \infty} n\left(\frac{t^2}{2n} + O\left(\frac{t^3}{n\sqrt{n}}\right)\right)} = e^{\frac{t^2}{2} + \lim_{n \rightarrow \infty} O\left(\frac{t^3}{\sqrt{n}}\right)} = e^{\frac{t^2}{2}}.$$

$$\left. \begin{array}{l} M'_{Y_1}(0) = E[Y_1] = 0 \\ M''_{Y_1}(0) = E[Y_1^2] = D^2[Y_1] = 1 \end{array} \right\} \Rightarrow M_{Y_1}(t) = 1 + 0 \cdot (t-0) + \frac{1}{2}(t-0)^2 + O(t^3)$$

by series expansion

Example: An astronomer measures the distance of a star from earth.

The values she gets are iid. random variables with expectation μ and variance 4 . She wants to estimate μ with the average of the measurements. How many measurements are necessary to make this estimate at least 0.5 lightyear accurate with probability at least 0.95?

X_n = the n^{th} measurement

Find $n \in \mathbb{N}$ for which $P\left(\left|\frac{X_1 + \dots + X_n}{n} - \mu\right| \leq 0.5\right) \geq 0.95$.

$$P\left(\left|\frac{X_1 + \dots + X_n}{n} - \mu\right| \leq 0.5\right) = P\left(\left|\frac{X_1 + \dots + X_n - n\mu}{\sqrt{n} \cdot 2}\right| \leq \frac{0.5\sqrt{n}}{2}\right) \underset{\text{CLT}}{\approx} \Phi\left(\frac{\sqrt{n}}{4}\right) - \Phi\left(-\frac{\sqrt{n}}{4}\right) \geq 0.95$$

$$\Rightarrow 0.95 \leq 2\Phi\left(\frac{\sqrt{n}}{4}\right) - 1 \Rightarrow 0.975 \leq \Phi\left(\frac{\sqrt{n}}{4}\right) \Rightarrow 1.96 \leq \frac{\sqrt{n}}{4} \Rightarrow 61.5 \leq n$$

$$\boxed{62 \leq n}$$

If we apply Chebisev's inequality instead

$$P(|\bar{X}_n - \mu| \leq 0.5) = 1 - P(|\bar{X}_n - \mu| > 0.5) \geq 1 - \underset{\text{Chebisev}}{\frac{4/n}{(0.5)^2}} = 1 - \frac{16}{n} \geq 0.95$$

$\Rightarrow \boxed{320 \leq n}$